SabiLens: A Multimodal AI Ecosystem for National Product Integrity

We are creating a verified multi-angle reference dataset using authentic manufacturer-sourced products. Each product is stored as a six-view embedding profile, and we compare incoming scans against that ground-truth baseline using similarity scoring and anomaly thresholds.

What reports the fake?

Our multi-signal fusion engine automatically generates a structured evidence file when authenticity confidence drops below threshold, which is then logged to the regulatory dashboard and optionally escalated with user consent.

Below is a **clear end-to-end architecture breakdown + 7-Day execution table**, explicitly split between:

- **AI Engineers (A1–A8)** → Models, pipelines, intelligence, geo-analytics
- **Developers (D1–D8)** → Mobile app, backend, APIs, database, dashboards, DevOps

Day 7 Must Show

During demo:

1. Scan authentic worn product → Authentic (High Confidence)
2. Scan counterfeit with typo → Suspicious / Fake
3. Show purchase channel selection
4. Dashboard shows geo cluster
5. Voice says, “Do not buy this product.”

AI ENGINEERS (A1-A4)

A1 - Visual & Damage Modelling Lead

What You're Building: The core image similarity engine + wear & tear detection.

Why It Matters: You are the system's "eyes." If your similarity scoring is weak, the entire authenticity decision collapses. Your damage detection prevents false positives when products are dented or scratched.

Impact on Demo: Your work proves the system can detect subtle visual differences while tolerating real-world packaging wear.

Day	Primary Task	Responsibilities	Deliverables
1	Reference Dataset Structuring	Organize 6-angle authentic product images; standardize resolution; define embedding architecture	Cleaned reference dataset + embedding plan
2	Embedding Pipeline	Implement CNN embedding extraction (ResNet/EfficientNet); generate embeddings for all reference images	Stored product embedding library
3	Region-Based Detection	Build logo/brand/NAFDAC region cropper; compute region embeddings	Region similarity scoring API
4	Damage Detection Layer	Build dent/scratch/blur detector; define damage score	Damage scoring module
5	Robustness Improvements	Add augmentation (blur, brightness, rotation); recalibrate similarity thresholds	Stable visual similarity engine
6	Optimization & Tuning	Reduce inference time; adjust thresholds for false positives	Optimized visual pipeline
7	Demo Simulation	Prepare 1 authentic worn product + 1 counterfeit test case	Demo-ready visual test scenarios

A2 - OCR & Linguistic Integrity Lead

What You're Building: Text extraction, typo detection, spelling validation, structural text checks.

Why It Matters: Counterfeiters often misspell brand names or alter batch formats. Even when visuals look correct, linguistic errors expose fakes.

Impact on Demo: You enable the "Panad0l vs Panadol" moment that impresses judges.

Day	Primary Task	Responsibilities	Deliverables
1	OCR Pipeline Setup	Integrate PaddleOCR; define text extraction schema	OCR extraction working locally
2	Metadata Extraction	Extract NAFDAC No., Batch, Expiry; normalize formats	Clean structured metadata JSON
3	Spelling Detection	Build custom dictionary for brands; detect typos & lookalikes	Linguistic anomaly score
4	Grammar & Layout Checks	Detect missing required text fields; detect malformed labels	Structural text validation module
5	Confidence Calibration	Weight OCR confidence vs anomaly signals	Text anomaly scoring logic
6	Integration Testing	Connect OCR outputs to fusion engine	Stable text validation API
7	Demo Cases	Prepare typo counterfeit example	Demo-ready spelling test

A3 - Greenbook RAG & Regulatory Lead

What You're Building: NAFDAC Greenbook ingestion and regulatory cross-verification engine.

Why It Matters: A valid-looking NAFDAC number means nothing if it belongs to another product category. You prevent regulatory mismatch fraud.

Impact on Demo: You power the moment where the system says- "Valid number, wrong product category."

Day	Primary Task	Responsibilities	Deliverables
1	Greenbook Structuring	Clean NAFDAC Greenbook; format into ingestion-ready dataset	Structured regulatory dataset
2	Vector DB Setup	Generate embeddings; load into ChromaDB	Searchable vector store
3	Retrieval Logic	Implement product-to-NAFDAC matching logic	Retrieval API
4	Categorical Alignment	Detect mismatch (e.g., milk vs tomato paste)	Regulatory mismatch engine
5	Confidence Scoring	Assign regulatory validity score	Regulatory score output
6	Integration	Connect regulatory score to fusion system	Live regulatory validation
7	Demo Prep	Create mismatch counterfeit scenario	Demo-ready regulatory case

A4 - Fusion, Intelligence & Monetization Architect

What You're Building: Convert multi-signal detection outputs into structured, severity-ranked intelligence, and design a role-based access system that enables regulatory oversight and subscription-based manufacturer analytics.

Why It Matters: You combine visual, linguistic, structural, and regulatory signals into one intelligent verdict. Without you, the system is just disconnected modules.

Impact on Demo: You produce the final; Authentic / Suspicious / Fake + Confidence Level.

You are the brain. The Fusion Engine feeds a centralized intelligence database with severity-ranked cases. Regulatory bodies receive full-spectrum access, while manufacturers access brand-specific analytics through a subscription-gated portal powered by role-based data filtering.

Day	**Primary Task**	**Responsibilities**	**Deliverables**
Day 1	Fusion Architecture Design	Design weighted scoring formula (visual + OCR + regulatory + linguistic + structure); define score thresholds	Fusion framework document
Day 2	Unified Intelligence Schema	Design centralized intelligence database schema; include case ID, brand tag, severity score, geo-tag, channel type	Intelligence DB schema (JSON structure)
Day 3	Authenticity Decision Engine	Implement final classification logic (Authentic / Suspicious / Fake + Confidence band)	Working classification module
Day 4	Severity & Risk Modelling	Build severity tagging (Low, Medium, High Risk); include damage-adjusted weighting logic	Severity scoring layer
Day 5	Role-Based Access Logic	Define access tiers: • Regulatory = full dataset • Manufacturer = brand-scoped dataset only Design brand-tag filtering mechanism	Role-based output schema
Day 6	Monetization Logic Structuring	Design subscription logic hooks (API gating); define paid analytics fields (trend reports, channel analytics, geo clusters)	Monetization-ready API structure
Day 7	Demo Intelligence Orchestration	Ensure flagged scan populates regulatory dashboard; ensure brand-scoped data appears only when filtered; prepare demo severity narrative	Live unified intelligence demo

AI DEVELOPERS (D1–D8)

D1 - Backend Lead (FastAPI Core)

What You're Building: The FastAPI backbone that connects mobile → AI → database.

Why It Matters: If backend fails, nothing works. You are the nervous system of the project.

Impact on Demo: Ensures scanning feels real-time and seamless.

Day	Primary Task	Responsibilities	Deliverables
1	Backend Setup	Initialize FastAPI; project structure	Running backend server
2	Image Upload API	Build image upload endpoint	Working upload API
3	AI Pipeline Trigger	Connect upload → AI pipeline	Async processing pipeline
4	Response Structuring	Return structured authenticity result	API response schema
5	Logging	Store evidence logs	Persistent scan logs
6	Optimization	Reduce latency; error handling	Stable backend
7	Deployment	Deploy final API	Live demo backend

D2 – Database & Storage Engineer

What You're Building: Secure evidence logging system (images, metadata, scores, purchase-channel, GPS).

Why It Matters: SabiLens is not just detection; it is regulatory intelligence. Without structured logs, there is no enforcement story.

Impact on Demo: Makes dashboard heatmap possible.

Day	Primary Task	Responsibilities	Deliverables
1	DB Schema	Design scan log schema	DB structure created
2	Image Storage	Configure cloud/local image storage	Image storage active
3	Evidence Logging	Store metadata + similarity scores	Complete evidence storage
4	Purchase-Channel Logging	Add channel attribution storage	Channel data saved
5	Query Optimization	Improve DB retrieval speed	Optimized queries
6	Backup Strategy	Create backup system	Recovery mechanism
7	Demo Stability	Final DB validation	Stable data persistence

D3 - Mobile App Lead

What You're Building: The user-facing experience.

Why It Matters: Judges see the mobile app first. If UX feels clunky, the AI strength won't matter.

Impact on Demo: Clean scan → instant result → voice feedback = wow factor.

Day	Primary Task	Responsibilities	Deliverables
1	App Setup	Initialize Flutter/React Native	Base mobile app
2	Camera Integration	Capture product image	Working camera feature
3	API Integration	Connect app to backend	Scan → result flow
4	Result Display UI	Show Authentic / Fake status	Visual decision screen
5	Purchase-Channel UI	Add vendor selection options	Channel selection screen
6	Voice Feedback UI	Integrate voice playback	Audio feedback working
7	Demo Polish	Improve UI & animations	Demo-ready mobile app

D4 – GPS & Geo Engineer

What You're Building: Mandatory location capture and geo-data structuring.

Why It Matters: Counterfeit detection without location is useless for enforcement. You turn a scan into actionable intelligence.

Impact on Demo: Enables live counterfeit heatmap.

Day	Primary Task	Responsibilities	Deliverables
1	GPS Setup	Enable mandatory GPS capture	GPS integrated
2	Location Logging	Store coordinates with scan	Location saved
3	Geo API	Create endpoint for geo data	Geo API ready
4	Data Formatting	Format for heatmap use	Geo dataset structured
5	Heatmap Backend Support	Provide geo clusters to dashboard	Geo clustering feed
6	Testing	Test coordinate accuracy	Reliable location capture
7	Demo Validation	Confirm live mapping works	Working geo demo

D5 – Dashboard Frontend

What You're Building: The regulatory and manufacturer dashboard.

Why It Matters: This proves SabiLens is bigger than a consumer app. It shows government and B2B scalability.

Impact on Demo: The heatmap visualization wins the room.

Day	Primary Task	Responsibilities	Deliverables
1	Admin Panel Setup	Create dashboard layout	Basic dashboard UI
2	Heatmap Integration	Integrate Mapbox/Leaflet	Live heatmap
3	Channel Filters	Add purchase-channel filters	Filter controls
4	Metrics Display	Add score summaries & stats	Analytics cards
5	Manufacturer View	Add product-based analytics	Brand panel
6	UI Refinement	Improve visuals & clarity	Clean dashboard
7	Demo Storyboard	Prepare dashboard demo flow	Demo-ready dashboard

D6 – Voice & UX Integration

What You're Building: Multilingual voice feedback and accessibility layer.

Why It Matters: Mama Bola must understand the result instantly. Literacy barriers are real.

Impact on Demo: When the app speaks in Yoruba or Pidgin, it humanizes the AI.

Day	Primary Task	Responsibilities	Deliverables
1	Audio Framework	Setup voice playback module	Audio module ready
2	Connect AI Output	Convert result to speech	Voice output working
3	Language Toggle	Add language selection	Language selector
4	UX Feedback	Add loading indicators	Better UX feedback
5	Accessibility	Improve readability & large fonts	Accessible UI
6	Latency Fix	Optimize voice timing	Smooth playback
7	Demo Prep	Validate voice scenario	Demo voice working

D7 - DevOps & Performance

What You're Building: Deployment, containerization, performance optimization.

Why It Matters: A crashing demo kills credibility. You ensure stability and speed.

Impact on Demo: Everything runs smoothly on presentation day.

Day	Primary Task	Responsibilities	Deliverables
1	Docker Setup	Containerize backend	Dockerized backend
2	Environment Config	Setup staging environment	Staging ready
3	Monitoring	Add logging & monitoring	Performance logs
4	Load Testing	Simulate scan loads	Load test results
5	Optimization	Reduce processing delay	Faster inference
6	Deployment Prep	Prepare demo server	Deployment-ready
7	Live Demo Support	Monitor during demo	Stable live system

D8 – QA & Integration Engineer

What You're Building: Full system testing, bug tracking, and edge-case validation.

Why It Matters: You prevent embarrassing live failures. You protect the team.

Impact on Demo: Confidence. Clean execution. No surprises.

Day	Primary Task	Responsibilities	Deliverables
1	Test Plan	Write full system test plan	QA checklist
2	Integration Tests	Test mobile → backend → AI	Integration report
3	Edge Case Testing	Test damaged packaging	Edge case report
4	False Positive Testing	Validate authenticity results	Calibration feedback
5	Bug Tracking	Log & track system bugs	Bug tracker
6	Final QA Sweep	Full system walkthrough	QA clearance
7	Demo Rehearsal	Simulate judge interaction	Demo stability report